

PH102

After Mid Semester Exams

by
Prof. Subhash Thota & Dr. Pankaj Kumar Mishra

Department of Physics, IIT Guwahati

Chapter-7

Energy in Magnetic Fields

Maxwell's Equations

Energy in Magnetic Fields

The work we must have to do against the **back emf** to get the current going.

This is a fixed amount, and it is recoverable: you get it back when the current is turned off. In the meantime it represents energy latent in the circuit; as we'll see in a moment, it can be regarded as energy stored in the magnetic field.

The work done on a unit charge, against the back emf, in one trip around the circuit is $-\varepsilon$ (the minus sign records the fact that this is the work done by you against the emf, not the work done by the emf). The amount of charge per unit time passing down the wire is 'I'. So the total work done per unit time is:

$$\frac{dW}{dt} = -\varepsilon I = L \frac{dI}{dt} I$$

Integrating over time

$$W = \frac{1}{2} L I^2$$

Here we start with zero current and build it up to a final value 'I'.

It depends only on the geometry of the loop and current

$$\Phi = \int_S \vec{B} \cdot \vec{da}$$

$$\Phi = \int_S \nabla \times \vec{A} \cdot \vec{da}$$

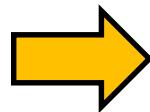
$$\Phi = \oint_c \vec{A} \cdot \vec{dl}$$

$$\Phi = LI$$

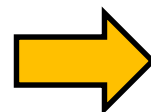
$$LI = \oint_c \vec{A} \cdot \vec{dl}$$

Hence

$$W = \frac{1}{2} LI^2$$



$$W = \frac{1}{2} I \oint \vec{A} \cdot \vec{dl}$$



$$W = \frac{1}{2} \oint \vec{A} \cdot \vec{I} dl$$

The generalization to volume currents is obvious:

$$W = \frac{1}{2} \oint_V \vec{A} \cdot \vec{J} d\tau$$

But we can do even better, and express W entirely in terms of the magnetic field:

The Ampere's law,

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

Let's eliminate $\vec{J}$:

$$W = \frac{1}{2\mu_0} \oint_V \vec{A} \cdot \nabla \times \vec{B} d\tau$$

Equation 1

The vector product rule:

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$$



$$\vec{A} \cdot (\nabla \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \nabla \cdot (\vec{A} \times \vec{B})$$

So

$$W = \frac{1}{2\mu_0} \oint_V \vec{A} \cdot \nabla \times \vec{B} d\tau$$

$$\vec{A} \cdot (\nabla \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \nabla \cdot (\vec{A} \times \vec{B})$$

Since

$$\nabla \times \vec{A} = \vec{B}$$

$$\vec{A} \cdot (\nabla \times \vec{B}) = \vec{B} \cdot (\vec{B}) - \nabla \cdot (\vec{A} \times \vec{B})$$

Substituting this value back to the equation 1

$$W = \frac{1}{2\mu_0} \oint_V \vec{B} \cdot (\vec{B}) - \nabla \cdot (\vec{A} \times \vec{B}) d\tau$$

$$W = \frac{1}{2\mu_0} \oint_V B^2 d\tau - \oint_V \nabla \cdot (\vec{A} \times \vec{B}) dl$$

$$W = \frac{1}{2\mu_0} \oint_V B^2 d\tau - \oint_S (\vec{A} \times \vec{B}) dl$$

if we agree to integrate over all space, then the surface integral goes to zero, and we are left with (total volume (∞) and no surface)

$$W = \frac{1}{2\mu_0} \int_{all\ space} B^2 d\tau$$

In view of this result, we say the energy is "stored in the magnetic field," in the amount $(B^2/2\mu_0)$ per unit volume.

In the light of this, it is extraordinary how similar the magnetic energy formulas are to their electrostatic counterparts:

$W = \frac{1}{2} \int (V\rho) d\tau$	$W = \frac{\epsilon_0}{2} \int E^2 d\tau$
Electrostatics	
Magneto-statics	
$W = \frac{1}{2} \int (\vec{A} \cdot \vec{j}) d\tau$	$W = \frac{1}{2\mu_0} \int B^2 d\tau$

7.3. Maxwell's Equations

Electrodynamics Before Maxwell

So far, we have encountered the following laws, specifying the divergence and curl of electric and magnetic fields:

Gauss's Law	$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$	1
No name	$\nabla \cdot \vec{B} = 0$	2
Faraday's Law	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	3
Ampere's Law	$\nabla \times \vec{B} = \mu_0 \vec{J}$	4

James Clerk Maxwell (1831–1879)



Formulate the classical theory of [electromagnetic radiation](#), bringing together for the first time Electricity, [Magnetism](#), and Light as manifestations of the same phenomenon.

[Maxwell's equations](#) for electromagnetism have been called the "second great unification in physics" after the first one realized by [Isaac Newton](#).

[University of Edinburgh](#)

Maxwell proved that the [Rings of Saturn](#) were made of numerous small particles.



Fatal inconsistency in these formulas

If we apply the divergence to Faraday's Law, everything works :

$$\nabla \cdot [\nabla \times \vec{E}] = -\frac{\partial(\nabla \cdot \vec{B})}{\partial t}$$

$$\nabla \cdot [\nabla \times \vec{E}] = 0$$

However, when we do the same thing to Ampere's Law, we get into trouble:

$$\nabla \cdot (\nabla \times \vec{B}) = \mu_0(\nabla \cdot \vec{J})$$

The left side must be zero, but the right side, in general, is not.

For steady currents, the divergence of $\vec{J}$ is zero, but evidently when we go beyond magneto-statics Ampere's law cannot be right.

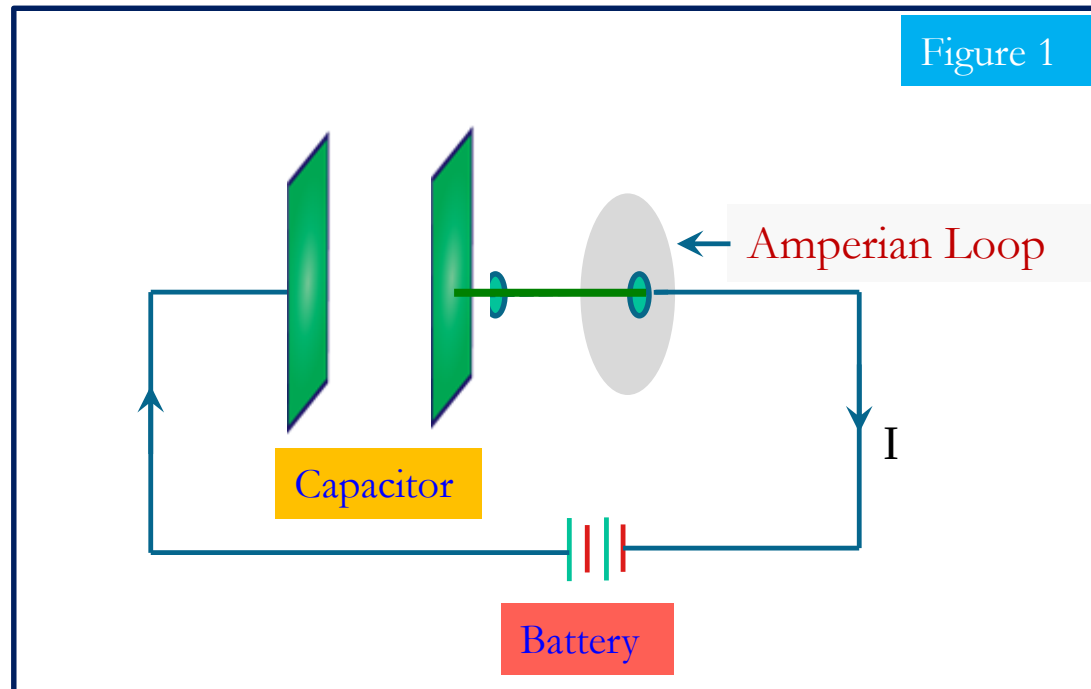
Another way to see that Ampere's law is bound to fail for non-steady currents.

Suppose we're in the process of charging up a capacitor (Figure 1)

In integral form, the Ampere's law reads

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

Applying it to the Amperian loop as shown in the diagram.



$$I_{\text{enclosed}} = \text{total } I \text{ passing through the loop}$$

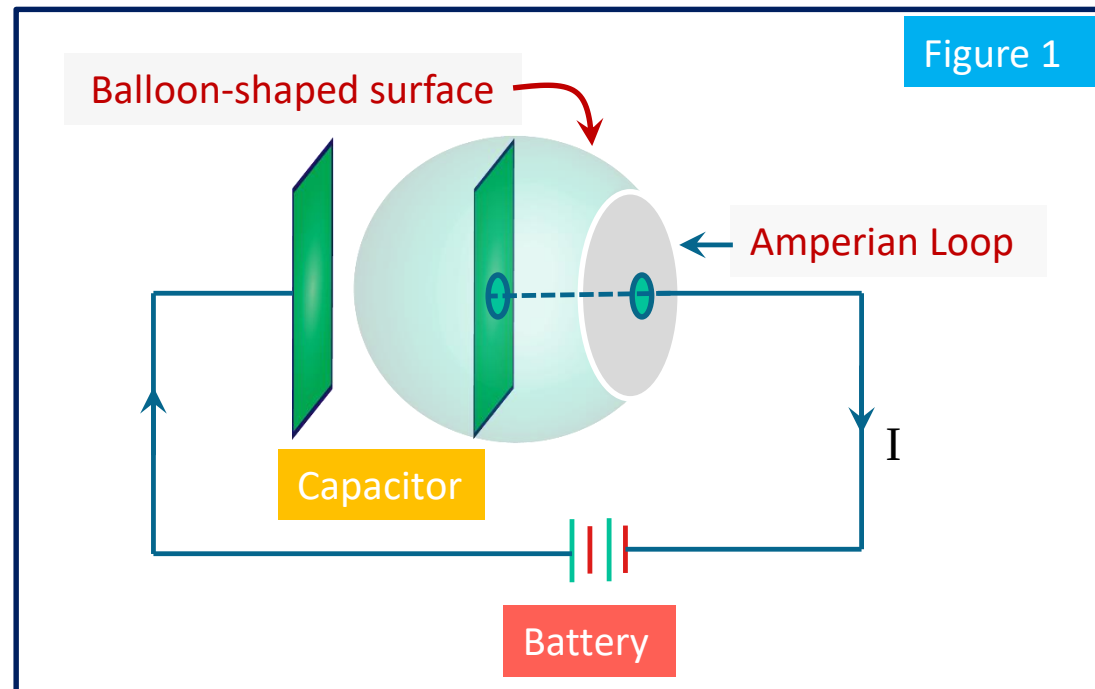
However, if we draw instead the balloon-shaped surface as shown in Figure 1.

$I_{\text{enclosed}} = 0$ No current passes through this surface, and $I_{\text{enc}} = 0$!

The conflict arises only when charge is piling up somewhere (in this case, on the capacitor plates).

For non-steady currents (such as this one) "the current enclosed by a loop" is an ill-defined notion, since it depends entirely on what surface you use.

We had no right to expect Ampere's law to hold outside of magneto-statics; after all, we derived it from the Biot-Savart law.



Ampere's Law (Maxwell's modification)

Using the continuity equation

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\nabla \cdot \vec{J} = -\frac{\partial}{\partial t} (\epsilon_0 \nabla \cdot \vec{E})$$

$$\nabla \cdot \vec{J} = -\nabla \cdot \left(\epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

Extra term is coming
in the divergence of J
So we have to this
overall J in Amperes
law

It might occur to you that if we were to combine $\epsilon_0 \frac{\partial \vec{E}}{\partial t}$ with J, in Ampere's law, it would be just right to kill off the extra divergence:

Equation 5

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Maxwell's term is hard to detect in ordinary electromagnetic experiments, where it must compete for recognition with J; that's why Faraday and the others never discovered it in the laboratory.

Such a modification changes nothing, as far as magnetostatics is concerned: when E is constant, we still have :

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

Maxwell's term has a certain aesthetic appeal: Just as a changing magnetic field induces an electric field

A changing electric field induces a magnetic field.

The real confirmation of Maxwell's theory came in 1888 with Hertz's experiments on electromagnetic waves.

Maxwell called his extra term the displacement current:

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Although it has nothing to do with current

Let's see now how the displacement current resolves the paradox of the charging capacitor

If the capacitor plates are very close together then the electric field between them is

$$E = \frac{\sigma}{\epsilon_0}$$

$$E = \frac{1}{\epsilon_0} \frac{Q}{A}$$

Here Q is the charge on the plate and A is its area.

Thus between the plates

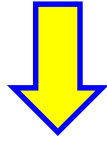
$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A} \frac{\partial Q}{\partial t}$$

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A} I$$

Equation 6

Now, Equation 5 reads, in integral form,

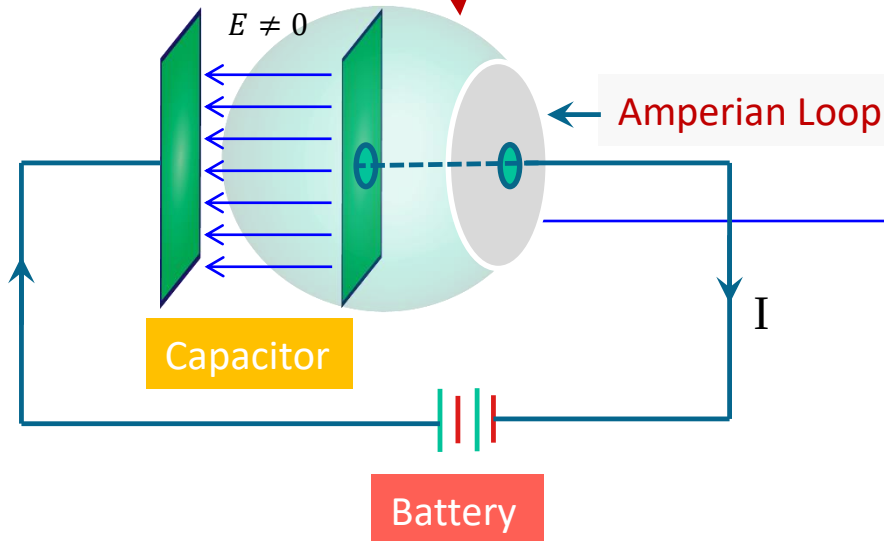
$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enclosed} + \mu_0 \epsilon_0 \int \left(\frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{a}$$

Figure 1

Balloon-shaped surface



If we choose the flat surface, then

$$E = 0$$

$$I_{enclosed} = I$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

If we choose the balloon surface

$$I_{enclosed} = 0$$

From Eq-6

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A} I$$

$$\int \left(\frac{\partial \vec{E}}{\partial t} \right) \cdot \vec{da} = \frac{I}{\epsilon_0}$$

$$\begin{aligned} \oint \vec{B} \cdot \vec{dl} &= 0 + \mu_0 \epsilon_0 \int \left(\frac{\partial \vec{E}}{\partial t} \right) \cdot \vec{da} \\ &= \mu_0 \epsilon_0 \frac{I}{\epsilon_0} = \mu_0 I \end{aligned}$$

Balloon-shaped surface

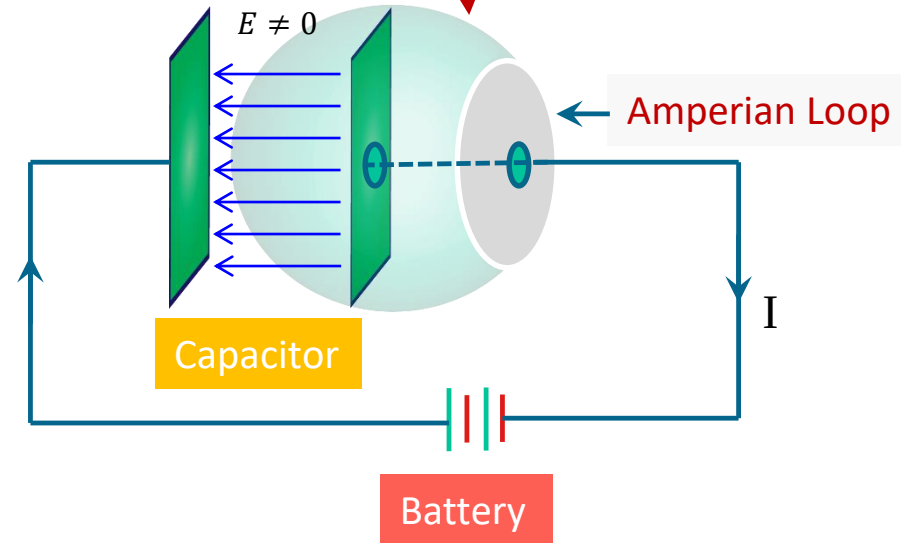
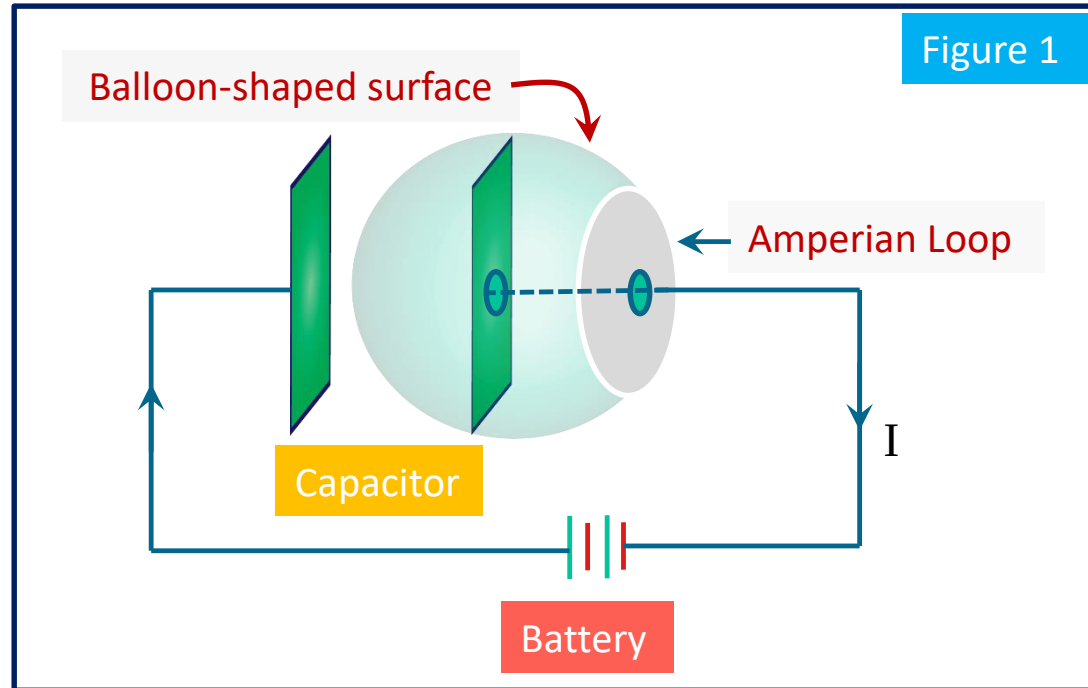
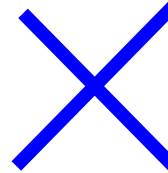


Figure 1

So we get the same answer for either surface, though in the first case it comes from the genuine current and in the second from the displacement current.

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 I_{enclosed} + \mu_0 \epsilon_0 \int \left(\frac{\partial \vec{E}}{\partial t} \right) \cdot \vec{da}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enclosed}$$



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enclosed} + \mu_0 \epsilon_0 \int \left(\frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{a}$$



When we go beyond Magnetostatics Ampere's Law can not be totally correct. Its should be modified as above with the second term.

Example 1: A fat wire of radius a , carries a constant current I , uniformly distributed over its cross section. A narrow gap in the wire, of width $w \ll a$, forms a parallel-plate capacitor. as shown in [Figure 2](#). Find the magnetic field in the gap, at a distance $s < a$ from the axis.

Solution: The displacement current density

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\vec{J}_d = \frac{\vec{I}}{A}$$

$$\vec{J}_d = \frac{I}{\pi a^2} \hat{z}$$

Drawing an “Amperian loop” at radius ‘ s ’,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{d-\text{enclosed}}$$

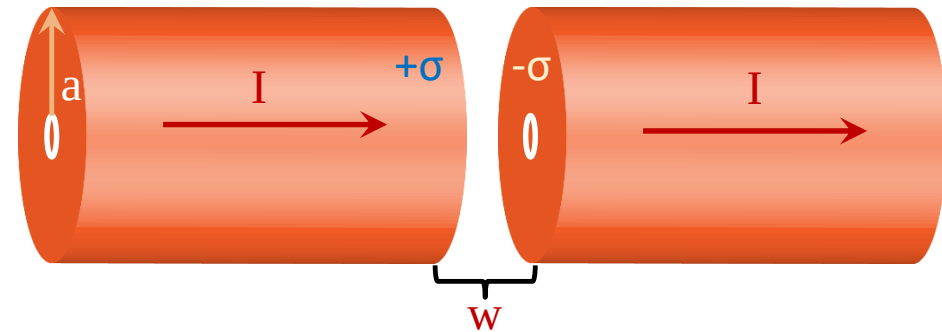
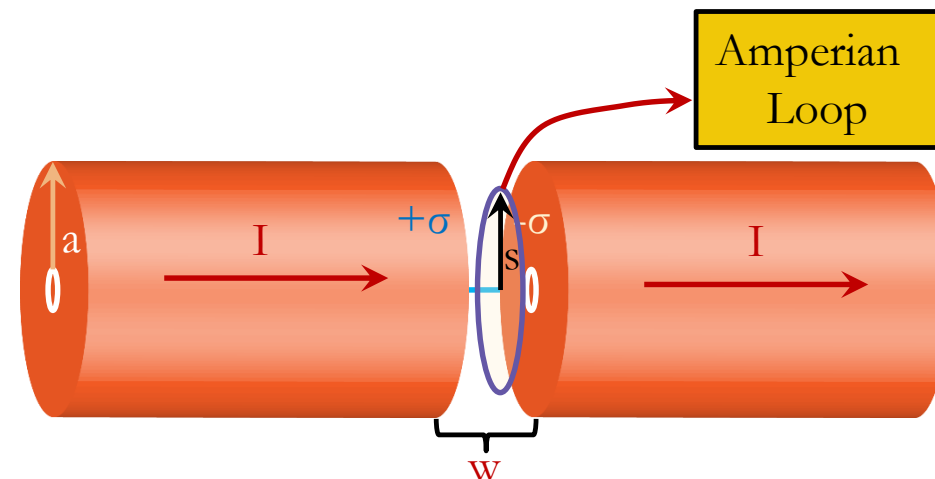


Figure 2

From Equation 6 (see slide # 21)



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{d-enclosed}$$

$$B \times 2\pi s = \mu_0 I_{d-enclosed}$$

$$= \mu_0 \frac{I}{\pi a^2} \pi s^2$$

$$= \mu_0 \frac{Is^2}{a^2}$$

$$B \times 2\pi s = \mu_0 \frac{Is^2}{a^2}$$

$$B = \frac{\mu_0 Is}{2\pi a^2} \hat{\phi}$$

Let's see now how the displacement current resolves the paradox of the charging capacitor

If the capacitor plates are very close together then the electric field between them is

$$E = \frac{\sigma}{\epsilon_0}$$

$$E = \frac{1}{\epsilon_0} \frac{Q}{A}$$

Here Q is the charge on the plate and A is its area.

Thus between the plates

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A} \frac{\partial Q}{\partial t}$$

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A} I$$

Equation 6

Maxwell's Equations

Gauss's Law	$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$	1
No name	$\nabla \cdot \vec{B} = 0$	2
Faraday's Law	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	3
Ampere's Law with Maxwell's Correction	$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$	5

Derivation of continuity relation using Maxwell's equation

Applying the divergence to equation number 5

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \cdot (\nabla \times \vec{B}) = \mu_0 (\nabla \cdot \vec{J}) + \mu_0 \epsilon_0 \frac{\partial (\nabla \cdot \vec{E})}{\partial t}$$

$$0 = \mu_0 (\nabla \cdot \vec{J}) + \mu_0 \epsilon_0 \frac{\partial (\nabla \cdot \vec{E})}{\partial t}$$

$$0 = (\nabla \cdot \vec{J}) + \epsilon_0 \frac{\partial (\nabla \cdot \vec{E})}{\partial t}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$



$$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

Continuity equation

Magnetic Charge

Symmetry about Maxwell's equations

It is particularly striking in free space, where ρ and J vanish:

First Pair

$$\nabla \cdot \vec{E} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Second Pair

If we replace $\mathbf{E}$ by $\mathbf{B}$ and $\mathbf{B}$ by $-\mu_0 \epsilon_0 \mathbf{E}$, the first pair of equations turning into the second, and vice versa.

This symmetry between E and B is spoiled, though, by the charge term in Gauss's law and the current term in Ampere's law.

Note that *if* magnetic charges ρ_m existed, then Maxwell's equations would become more “symmetrical”.

If magnetic charges ρ_m existed!!

$\nabla \cdot \vec{E} = \frac{\rho_e}{\epsilon_0}$	$\nabla \times \vec{E} = -\mu_0 J_m - \frac{\partial \vec{B}}{\partial t}$
$\nabla \cdot \vec{B} = \mu_0 \rho_m$	$\nabla \times \vec{B} = \mu_0 J_e + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

Then ρ_m would represent the density of magnetic "charge," and ρ_e the density of electric charge; J_m would be the current of magnetic charge, and J_e the current of electric charge. Both charges would be conserved:

$$\nabla \cdot \vec{J}_m + \frac{\partial \rho_m}{\partial t} = 0$$

$$\nabla \cdot \vec{J}_e + \frac{\partial \rho_e}{\partial t} = 0$$

In a sense, Maxwell's equations beg for magnetic charge to exist; it would fit in so nicely. And yet, in spite of a diligent search, no one has ever found any.

Maxwell's Equations in Matter

If we are working with materials that are subject to electric and magnetic polarization there is a more convenient way to write them.

For inside polarized matter there will be accumulations of "bound" charge and current over which we exert no direct control.

If we reformulate Maxwell's equations in such a way as to make explicit reference only to those sources we control directly: the "free" charges and currents.

We know from the static case of Polarization $\mathbf{P}$; it produces a bound charge density.

$$\rho_b = -\nabla \cdot \vec{P}$$

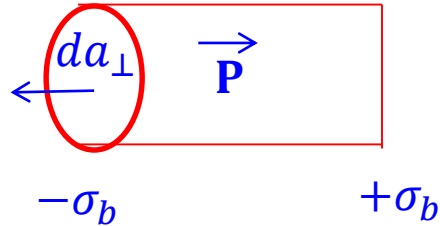
Similarly for Magnetic case:

$$J_b = \nabla \times \vec{M}$$

There is a one new feature to consider in the non-static case (dynamic):

The polarization induced a charge density

$$\sigma_b = P \cdot \hat{n}$$



$$\sigma_b = P$$

At one end

$$-\sigma_b = P$$

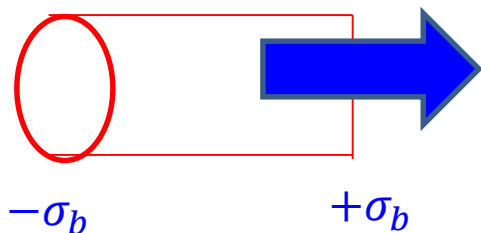
At other end

If P now increases a bit, the charge on each end increases accordingly. Giving a net current

$$dI = \frac{\partial Q_b}{\partial t} = \frac{\partial \sigma_b}{\partial t} da_{\perp} = \frac{\partial P}{\partial t} da_{\perp}$$

The current density, therefore, is

$$\mathbf{J}_P = \frac{\partial P}{\partial t}$$



Moving (Dynamic)

Polarization current results from
Dynamic case of Bound charges

Difference between bound current and Polarization current

$$J_b = \nabla \times \vec{M}$$

Bound
Current

- This is associated with magnetization of the material and involves the spin and orbital motion of electrons.

Polarization
Current

- This is the result of the linear motion of charge when the electric polarization changes.

$$J_P = \frac{\partial P}{\partial t}$$

$$\sigma_b = P$$

If $\vec{P}$ points to the right and is increasing, then each plus charge moves a bit to the right and each minus charge to the left; the cumulative effect is the polarization current $\vec{J}_P$.

Let's check the continuity equation for Polarization current

$$\vec{J}_P = \frac{\partial \vec{P}}{\partial t}$$

After taking the divergence

$$\nabla \cdot \vec{J}_P = \frac{\partial (\nabla \cdot \vec{P})}{\partial t}$$

$$\nabla \cdot \vec{P} = -\rho_b$$

$$\nabla \cdot \vec{J}_P = -\frac{\partial \rho_b}{\partial t}$$

Hence, the continuity equation is satisfied

The total charge density can be separated into two parts

$$\rho = \rho_f + \rho_b$$

$$\rho = \rho_f - \nabla \cdot \vec{P}$$

and the current density into three parts:

$$\vec{J} = \vec{J}_f + \vec{J}_b + \vec{J}_P$$

$$\vec{J} = \vec{J}_f + \nabla \times \vec{M} + \frac{\partial \vec{P}}{\partial t}$$

Gauss's law can now be written as

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$



$$\nabla \cdot \vec{E} = \frac{1}{\epsilon_0} (\rho_f - \nabla \cdot \vec{P})$$

$$\nabla \cdot \vec{E} = \frac{1}{\epsilon_0} (\rho_f - \nabla \cdot \vec{P})$$

$$\nabla \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho_f$$



$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\nabla \cdot \vec{D} = \rho_f$$

1st Maxwell's
Equation in Matter

Meanwhile, Ampere's law (with Maxwell's term) becomes

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 (\vec{J}_f + \vec{J}_b + \vec{J}_P) + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J}_f + \nabla \times \vec{M} + \frac{\partial \vec{P}}{\partial t} \right) + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J}_f + \nabla \times \vec{M} + \frac{\partial \vec{P}}{\partial t} \right) + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \frac{\vec{B}}{\mu_0} - (\nabla \times \vec{M}) = \left(\vec{J}_f + \frac{\partial \vec{P}}{\partial t} \right) + \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J}_f + \frac{\partial}{\partial t} (\vec{P} + \epsilon_0 \vec{E})$$

$$\nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

4th Maxwell's
Equation in Matter

Here

$$\vec{H} = \frac{1}{\mu_0} (\vec{B} - \vec{M})$$

Faraday's law and $\nabla \cdot \vec{B} = 0$ are not affected by our separation of charge and current into free and bound parts, since they do not involve ρ or $\mathbf{J}$.

Maxwell's equations (In terms of free charges and currents)

1	$\nabla \cdot \vec{D} = \rho_f$	3	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
2	$\nabla \cdot \vec{B} = 0$	4	$\nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$

Here, 'D' and 'H' depends on the nature of the material

For linear media

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$



$$\vec{D} = \epsilon \vec{E}$$

Here

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

Displacement Current

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$

$$\vec{M} = \chi_m \vec{H}$$



$$\vec{B} = \frac{1}{\mu} \vec{H}$$

$$\mu = \mu_0 (1 + \chi_m)$$

Example 2: Sea water at frequency ' ν ' = 4×10^8 Hz has permittivity $\epsilon = 81 \epsilon_0$, permeability $\mu = \mu_0$, and resistivity $\rho = 0.23 \Omega\text{-m}$. What is the ratio of conduction current to displacement current?

Solution 2: Consider a parallel-plate capacitor immersed in sea water and driven by a voltage $V_0 \cos(2\pi\nu t)$

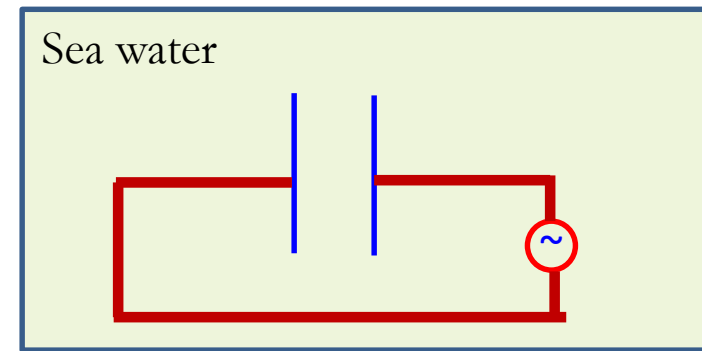
Electric field

$$E = \frac{V}{d}$$

Conduction current density

$$J_c = \sigma E$$

$$= \frac{E}{\rho}$$



$$J_c = \frac{V}{\rho d}$$

$$J_c = \frac{V_o}{\rho d}$$

Considering only Amplitude

Displacement current density

$$J_d = \epsilon \frac{\partial E}{\partial t}$$

$$J_d = \epsilon \frac{\partial E}{\partial t}$$

$$= \frac{\epsilon}{d} \frac{\partial}{\partial t} (V_0 \cos(2\pi\nu t))$$

$$J_d = \frac{\epsilon}{d} (-2\pi\nu) V_0 \sin(2\pi\nu t)$$

Considering only Amplitude



$$J_d = \frac{\epsilon}{d} (-2\pi\nu) V_0$$

Ratio of conduction current to displacement current =

$$\frac{J_c}{J_D}$$

$$J_c = \frac{V}{\rho d}$$

$$V_0 \cos(2\pi\nu t)$$

$$\epsilon = 81 \epsilon_0$$

$$\rho = 0.23 \Omega\text{-m}$$

$$\mu = \mu_0$$

$$\nu = 4 \times 10^8 \text{ Hz}$$

$$= \frac{V_0}{\rho d} \frac{d}{\epsilon (-2\pi\nu) V_0}$$

$$= -\frac{1}{2\pi\epsilon\nu\rho}$$

$$\frac{J_c}{J_D} = 2.41$$

Considering only Amplitude

Few Examples

Example: A long coaxial cable carries current 'I' (the current flows down the surface of the inner cylinder, radius 'a', and back along the outer cylinder, radius 'b') as shown in Figure 14. Find the magnetic energy stored in a section of length 'l'.

Solution: According to Ampere's law, the field between the cylinders is

$$\vec{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$$

Elsewhere, the field is zero. Thus, the energy per unit volume is

$$W = \frac{1}{2\mu_0} \int B^2 d\tau$$

$$u = W/\text{Vol} = \text{Energy per unit volume} = \frac{B^2}{2\mu_0}$$

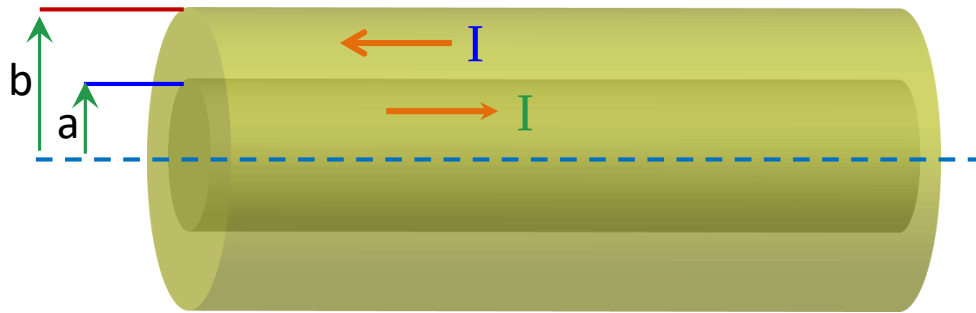


Figure 14

Energy density

$$u = \frac{\left(\frac{\mu_0 I}{2\pi s}\right)^2}{2\mu_0}$$

$$u = \frac{\mu_0 I^2}{8\pi^2 s^2}$$

The energy in a cylindrical shell of length l , radius s , and thickness ds , then, is

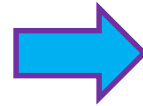
Volume of shell having thickness ds
 $dV = 2\pi l s ds$

$$dW = \frac{\mu_0 I^2}{8\pi^2 s^2} 2\pi l s ds$$

$$dW = \frac{\mu_0 I^2 l}{4\pi s} ds$$

Integrating from a to b

$$W = \int_a^b \frac{\mu_0 I^2 l}{4\pi s} ds$$



$$W = \frac{\mu_0 I^2 l}{4\pi} \ln\left(\frac{b}{a}\right)$$

Comparing with

$$W = \frac{1}{2} L I^2$$

$$L = \frac{\mu_0 l}{2\pi} \ln\left(\frac{b}{a}\right)$$

Example 6: A short solenoid (length l and radius a , with n_1 turns per unit length) lies on the axis of a very long solenoid (radius b , n_2 turns per unit length) as shown in Figure 8. Current I flows in the short solenoid. What is the flux through the long solenoid?

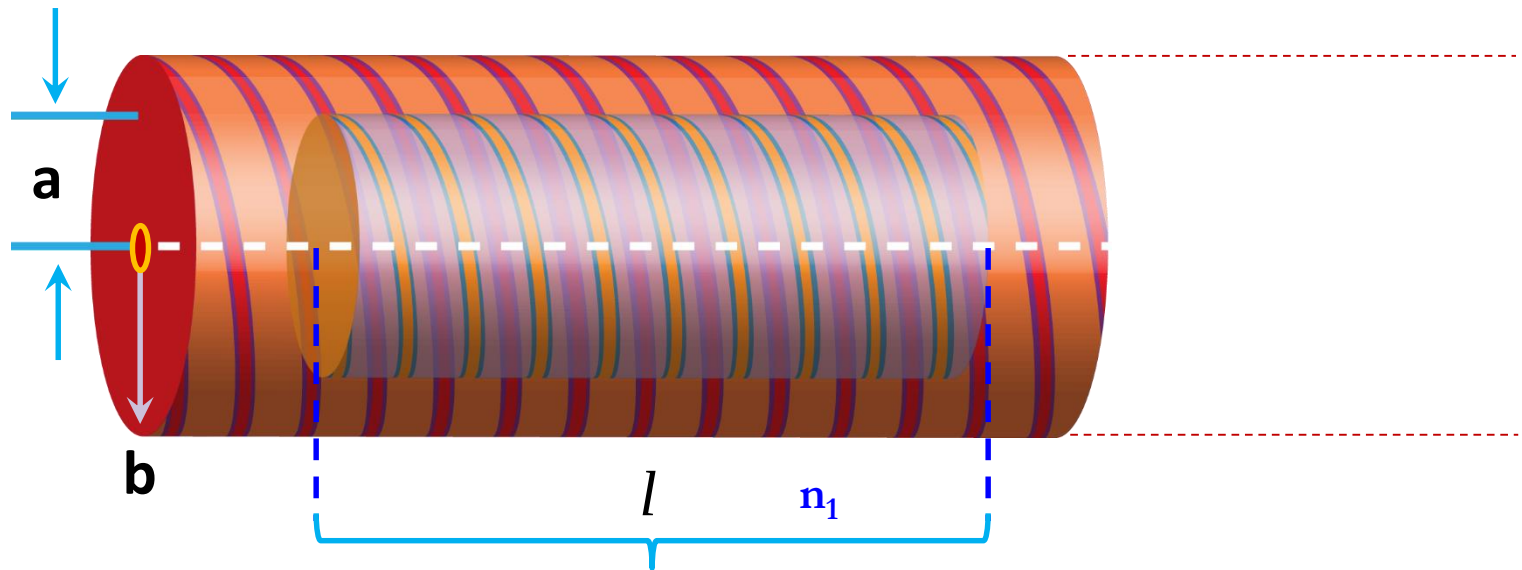


Figure 10

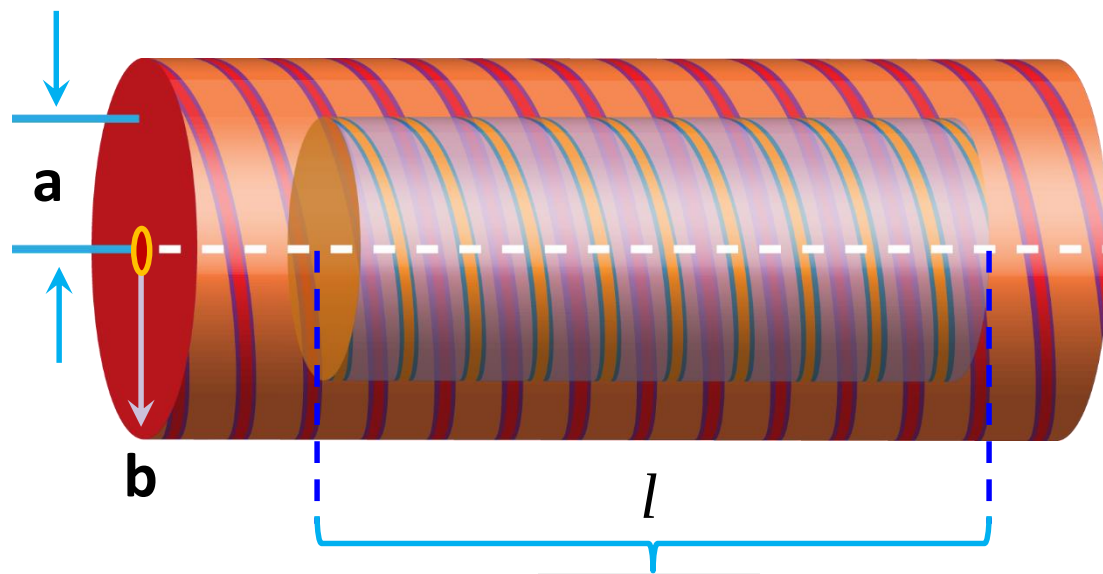


Figure 10

Solution: Since the inner solenoid is short, it has a very complicated field; moreover, it puts a different amount of flux through each turn of the outer solenoid. It would be a miserable task to compute the total flux this way.

However, if we exploit the equality of the mutual inductances, the problem becomes very easy.

Trick

If we look at the reverse situation: run the current ' I ' through the outer solenoid, and calculate the flux through the inner one.

The field inside the long solenoid is constant:

$$B = \mu_0 n_2 I$$

So the flux through a single loop of the short solenoid is

$$B\pi a^2 = \mu_0 n_2 I \pi a^2$$

There are $n_1 l$ turns in all, so the total flux through the inner solenoid is

$$\Phi = \mu_0 n_1 n_2 l \pi a^2 I$$

This is also the flux a current 'I' in the short solenoid would put through the long one, which is what we set out to find. Incidentally, the mutual inductance, in this case, is

$$\Phi_2 = M I_1$$

$$M = \mu_0 n_1 n_2 l \pi a^2$$

Example

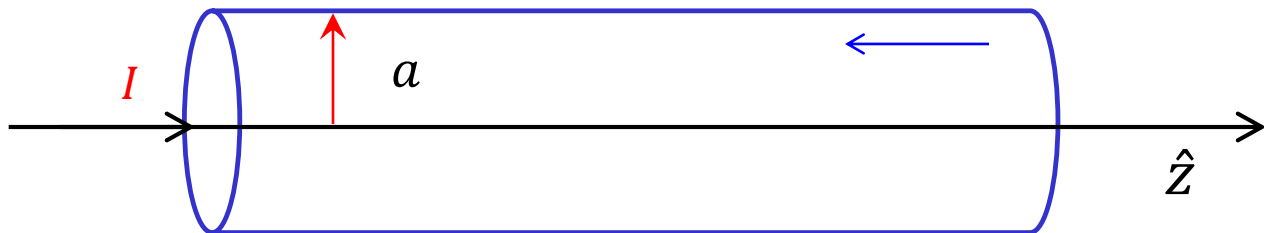
An alternating current $I = I_0 \cos(\omega t)$ flows down a long straight wire and returns along a coaxial conducting tube of radius a . And induced electric field is

$$\vec{E}(s, t) = \frac{\mu_0 I_0 \omega}{2\pi} \sin(\omega t) \ln\left(\frac{a}{s}\right) \hat{z}$$

- (a) Find the displacement current density J_d
- (b) Integrate it to get the total displacement current,

$$I_d = \int J_d \cdot d\vec{a}$$

- (c) Compare I_d and I . (What's their ratio?) If the outer cylinder is 2 mm in diameter, how high would the frequency have to be, for I_d to be 1% of I ?



Solution :

(a) Displacement current:

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$



$$\vec{J}_d = \frac{\mu_0 \epsilon_0 I_0 \omega^2}{2\pi} \cos(\omega t) \ln\left(\frac{a}{s}\right) \hat{z}$$

(b) Total displacement current:

$$I_d = \int \vec{J}_d \cdot d\vec{a}$$

$$I_d = \frac{\mu_0 \epsilon_0 I_0 \omega^2}{2\pi} \cos(\omega t) \int_0^a \ln\left(\frac{a}{s}\right) (2\pi s ds)$$

$$I_d = \mu_0 \epsilon_0 I_0 \omega^2 \cos(\omega t) \int_0^a (s [\ln a - \ln s]) ds$$



$$I_d = \frac{\mu_0 \epsilon_0 I_0 a^2 \omega^2}{4} \cos(\omega t)$$

(c)

$$I_d = \frac{\mu_0 \epsilon_0 I_0 a^2 \omega^2}{4} \cos(\omega t)$$

Now , $I = I_0 \cos(\omega t)$

$$\therefore I_d = \frac{\mu_0 \epsilon_0 I a^2 \omega^2}{4}$$



$$\frac{I_d}{I} = \frac{\mu_0 \epsilon_0 a^2 \omega^2}{4}$$

Since , $\mu_0 \epsilon_0 = 1/c^2$

$$\frac{I_d}{I} = \frac{a^2 \omega^2}{4c^2}$$

Given , $a = 1 \text{ mm} = 10^{-3} \text{ m}$
 I_d to be 1% of I : $\frac{I_d}{I} = \frac{1}{100}$

 $\omega = \frac{2c}{a} \left(\frac{I_d}{I} \right)^{1/2}$

$\omega = 6 \times 10^{10} / \text{s}$



$\nu = \frac{\omega}{2\pi} \approx 10^{10} \text{ Hz}$

This is in the microwave region.